

A04 AI in Fraud Detection Case Study

Financial institutions like banks use Artificial Intelligence to help detect fraud. Fraud is when someone tries to steal money or use someone else's account in a dishonest way. Since banks handle millions of transactions every day, it is impossible for humans to check everything. That is why AI is very important.

AI helps banks find fraud faster and more accurately. It looks for unusual activity in transactions and can stop fraud before it causes big problems. This helps protect both the bank and its customers.

Banks use machine learning and data analysis tools to detect fraud. Machine learning is when a computer learns from past data to make decisions.

For example, the system studies a customer's normal spending habits. If something unusual happens, like a large purchase in another country, the system will flag it as possible fraud.

Some banks also use real-time AI systems. These systems check transactions instantly and give them a risk score. If the score is too high, the transaction may be blocked or reviewed.

One example is how big banks like HSBC and Danske Bank use AI systems to monitor transactions. These systems help detect suspicious activity faster than older systems that only used fixed rules.

Instead of waiting for fraud to happen, AI can catch patterns early and prevent losses.

AI has many benefits in fraud detection:

- It works very fast and can check thousands of transactions in seconds
- It helps reduce financial losses from fraud
- It improves security for customers
- It saves time for employees because they don't have to check everything manually
- It can detect patterns that humans might miss

Even though AI is helpful, there are still some problems:

- Sometimes it flags normal transactions by mistake (false positives)
- It needs a lot of good data to work properly
- Poor or biased data can lead to bad results
- There are privacy concerns because sensitive financial data is used
- It can be expensive to build and maintain AI systems

AI is very useful for fraud detection in financial institutions. It helps banks catch fraud faster, reduce losses, and protect customers. However, it is not perfect. There are still challenges like false alerts and privacy concerns.

